

Safe and Interpretable Human-Like Planning With Transformer-Based Deep Inverse Reinforcement Learning for Autonomous Driving

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Abstract—Human-like decision-making and planning are crucial for advancing the decision-making level of autonomous driving and increasing acceptance in the autonomous vehicle market, as well as for achieving data closed loop for autonomous driving. However, human-like decision-making and planning methods still face challenges in safety and interpretability, particularly in multi-vehicle interaction scenarios. In light of this, this paper proposes an interpretable human-like decision-making and planning method with Transformer-based deep inverse reinforcement learning. The proposed method employs a Transformer encoder to extract features from the scenario and determine the attention assigned by the ego vehicle to each traffic vehicle, thereby improving the interpretability of planning outcomes. Furthermore, for improved safety in planning, the model is trained on both positive and negative expert demonstrations. The experimental results show that the proposed method enhances model safety while maintaining imitation levels compared to conventional methods. Additionally, the attention allocation results closely align with those of human drivers, indicating the model's ability to elucidate the importance of each traffic vehicle for decision-making and planning, thereby improving interpretability. Therefore, the proposed method not only ensures high levels of imitation and safety but also enhances interpretability by providing accurate attention allocation results for decision-making and planning.

Note to Practitioners—This paper presents a method for enhancing the planning of autonomous vehicles by making it more interpretable and safer. Using Transformer-based deep reinforcement learning, the approach improves clarity by showing how the vehicle prioritizes other traffic participants and learning from both positive and negative examples. This not only enhances

safety and decision accuracy but also provides insights into the vehicle's reasoning process, which is crucial for debugging and increasing user trust. Future work could focus on adapting this method for even more complex driving scenarios.

Index Terms—Autonomous driving, decision-making, planning, deep inverse reinforcement learning, interpretability.

I. INTRODUCTION

THE current autonomous driving technology still faces numerous challenges. Firstly, there remains a gap between the driving proficiency of autonomous vehicles (AVs) and human drivers, particularly in complex scenarios. Additionally, disparities in driving styles between AVs and human drivers contribute significantly to high takeover rates by passengers, traffic disruptions, and exacerbate the lack of trust in autonomous driving among passengers and other traffic participants [1]. The human-like decision-making and planning approach based on human driving data involves learning driving decisions and planning strategies from human driving data, thereby mimicking human driving styles. Consequently, human-like decision-making and planning are crucial for enhancing the performance of decision-making and planning and increasing market acceptance of AVs. Furthermore, closed-loop data is vital for the self-evolution of AVs, and human-like decision-making and planning methods constitute a crucial component in achieving closed-loop data for autonomous driving.

The theoretical foundation of human-like decision-making and planning methods stems from imitation learning (IL). IL aims to utilize human demonstration data to train agents, enabling them to mimic human behavior [2]. Human-like decision-making and planning methods involve training AVs using human driving data, thereby endowing them with driving proficiency and styles similar to human drivers. IL primarily consists of two methods: behavior cloning (BC) and inverse reinforcement learning (IRL). Consequently, human-like decision-making and planning methods can also be categorized into BC-based and IRL-based approaches. BC is essentially a form of supervised learning that utilizes expert demonstration data to train models, establishing a mapping from “states” to “actions” [3]. While BC superficially mimics the observable driving behaviors of human drivers, it fails to capture the underlying reasons behind these behaviors. In contrast, inverse reinforcement learning (IRL) aims to uncover the

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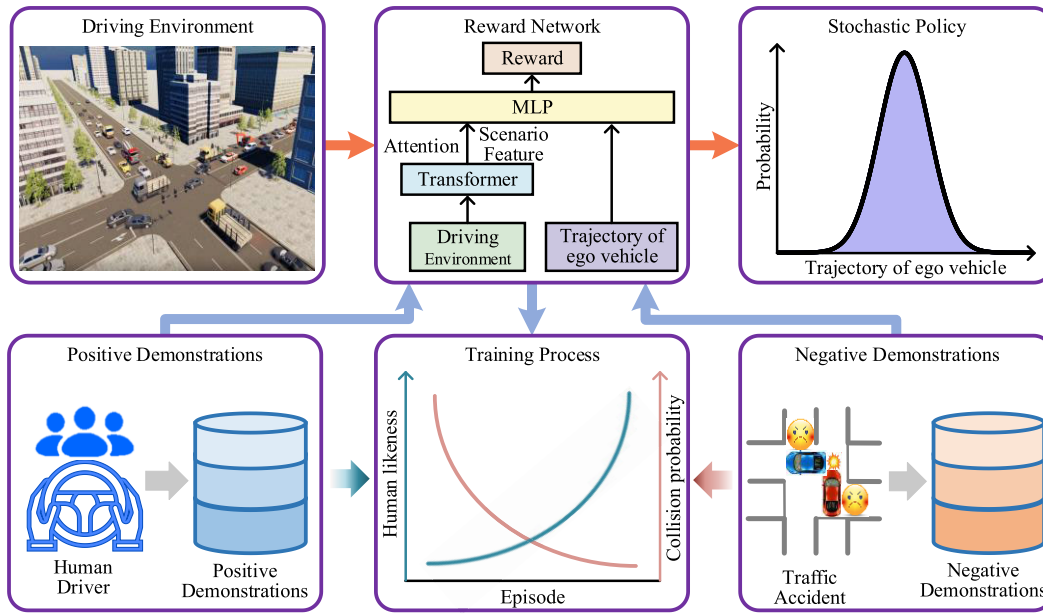


Fig. 1. Interpretable human-like decision-making and planning with transformer-based deep inverse reinforcement learning. The orange arrows depict the inference process, while the blue arrows illustrate the training process.

intrinsic driving intentions from the data. In IRL, a reward function is used to mathematically represent this intention, which serves as the fundamental reason behind the observed driving behaviors. By learning the more essential driving intentions embedded in behavior, IRL offers better generalization capabilities and avoids the causal confusion often encountered in BC. Traditional IRL methods based on manually extracted driving features suffer from limited fitting capabilities, inaccuracies in feature extraction, and an inability to account for contextual scenario information. In contrast, deep inverse reinforcement learning (DIRL) addresses these shortcomings by leveraging the representation power of deep learning. However, like other data-driven approaches, DIRL still faces significant challenges in ensuring both safety and interpretability.

The ability to pursue benefits and avoid harm is a fundamental aspect of human intelligence and a core advantage in human driving. In existing learning-based autonomous driving methods, human driver behavior is often considered the optimal driving policy, and the learning process aims to approximate this policy over time. However, these methods typically focus on benefit-seeking and neglect the ability to avoid harm, which leads to significant safety challenges, especially in hazardous scenarios. In such scenarios, human driving behavior is often very close to collision zones. In contrast, learning-based methods are prone to entering these zones due to convergence errors and a lack of collision zone awareness. Additionally, extracting accurate scenario features is essential for autonomous vehicles to make safe decisions. In complex scenarios, rule-based scenario feature extraction methods reduce scenario information by selecting important targets. However, this approach can result in information loss or incorrect selection of important targets. On the other hand, convolutional neural network-based methods preserve all scenario information but lack interpretability. Interpretability is critical in autonomous driving, as it allows engineers and researchers to test, debug,

and improve the system more efficiently. It also ensures performance guarantees from a societal perspective, increases user trust, and promotes public acceptance. Therefore, identifying collision areas and extracting accurate, interpretable scenario features are key to overcoming the safety and interpretability challenges in human-like autonomous driving.

To address the aforementioned challenges, a Safe and Interpretable Human-like Decision-Making and Planning method using Transformer-Based Deep Inverse Reinforcement Learning is proposed, as illustrated in Fig. 1. This approach restructures the objective function and optimizes the training framework of deep inverse reinforcement learning, enabling the model to learn from two distinct types of driving data: positive demonstrations, which provide valuable examples for learning, and negative demonstrations, which highlight behaviors to avoid. As a result, the proposed method not only learns advantageous positions in the planning space from expert positive demonstrations but also identifies collision zones in the planning process from negative demonstrations. Furthermore, by integrating a Transformer Encoder as a core component of the reward network in deep inverse reinforcement learning, the method effectively addresses the challenge of extracting scenario features. Leveraging the attention mechanism of the Transformer Encoder, the proposed method computes attention weights allocated by the ego vehicle to surrounding traffic participants, thereby providing clear interpretability for the planning outcomes.

The main contributions and innovations of this paper are as follows:

(1) To address the challenges of extracting scenario features and lack of interpretability in human-like decision-making and planning methods based on DIRL, we propose an interpretable human-like planning approach with Transformer-based DIRL. By integrating the Transformer encoder as a crucial component of the reward network, our proposed

method resolves the difficulty in extracting scenario features. Moreover, our proposed method computes attention allocated by the ego vehicle to surrounding traffic vehicles, providing interpretability to the planned trajectory.

(2) To enhance the safety of the planned trajectory, our proposed method, decision-making and planning method with PN-TIRL, not only learn from positive demonstrations, as in traditional deep inverse reinforcement learning, but also from negative demonstrations. It can derive insights from negative demonstrations, thereby more effectively avoiding collisions.

(3) We develop lateral and longitudinal trajectory tracking algorithms and establish hardware-in-the-loop experimental platforms. Through hardware-in-the-loop experiments, we validate the imitation, safety, and interpretability of the proposed method.

II. RELATED WORK

A. Human-Like Autonomous Driving

The methods for achieving human-like autonomous driving can be broadly categorized into behavior cloning-based approaches and inverse reinforcement learning-based approaches. The earliest application of BC in autonomous driving can be traced back to 1988 [4]. In 2016, NVIDIA introduced a pioneering architecture that employed convolutional neural networks (CNN) to establish the mapping between visual signals and steering commands [5]. Codevilla et al. propose conditional imitation learning, where the model's input is also visual signals, but it simultaneously outputs lateral steering angles and longitudinal acceleration commands [6]. In addition to directly outputting acceleration and steering control commands, some behavior cloning-based methods also output trajectories of autonomous vehicles [7], [8], [9]. BC achieves human-like autonomous driving by directly mimicking the mapping between states and actions in expert data. Although this approach spares the trouble of the hand-engineered reward functions or loss functions, it directly fits the mapping from states to actions without comprehending the reasons behind the actions taken by experts, leading to causal confusion and poor generalization. Furthermore, without sufficient edge case scenarios in the training data, BC-based methods struggle to learn human drivers' behavior in these situations, reducing their safety.

IRL is also a form of IL algorithm, introduced later than BC, with the aim of avoiding potential issues arising from the mere imitation of expert behavior without delving into the underlying reasons behind it [10], [11]. In contrast to BC, IRL does not simply learn the mapping from states to actions; rather, it infers the underlying reasons behind expert behavior based on expert behavior, which are mathematically represented as reward functions guiding these expert behaviors [12], [13], [14]. Compared to BC, IRL avoids causal confounding and exhibits stronger generalization. The term "inverse reinforcement learning" stems from its learning process being the opposite of reinforcement learning. While reinforcement learning relies on predefined reward functions to learn optimal strategies, IRL uncovers the reward functions guiding these expert strategies based on expert demonstration [15], [16], [17].

IRL is divided into the maximum margin-based method and the maximum entropy-based method. maximum margin-based IRL infers the expert's reward function by maximizing the decision margin probability [18], [19], [20], [21]. However, maximum margin-based IRL often leads to ambiguity, as many different reward functions can result in the same expert policy. In such cases, the learned reward function often exhibits random preferences. To address the ambiguity issue of reward functions in maximum margin-based IRL, Ziebart et al. proposed the maximum entropy-based IRL, which utilizes the maximum entropy principle to overcome the ambiguity of reward functions [22]. The maximum entropy-based IRL regards the expert policy as a probability model. In probability theory, "entropy" measures uncertainty, with higher uncertainty corresponding to higher entropy. The maximum entropy principle is used for learning probability models, asserting that among all probability models satisfying all constraints, the model with the maximum entropy is considered optimal because it does not make subjective assumptions about the unknown.

In maximum entropy-based IRL methods, feature-based IRL is a commonly employed approach. Feature-based IRL assumes that the expert's reward function can be represented as a linear function of multiple features. The training process of IRL involves determining the weight relationships between features. In the field of autonomous driving, features in feature-based IRL refer to the mathematical representations of driving subgoals of the autonomous vehicle. The decision-making and planning methods in feature-based IRL learn the reward function of human drivers from human driving data, which involves learning the weights of various driving subgoals [23], [24], [25]. It is evident that driving features play a crucial role in feature-based IRL methods. Naumann et al. extract driving features related to comfort, safety, driving efficiency, traffic regulations, and interactions with other traffic participants [26]. Sun et al. introduce driving features representing the level of politeness to make driving behavior more human-like [27], [28]. Due to the concise form and fewer parameters of the reward function, feature-based IRL is easy to train. However, it also suffers from the following issues: (1) Hand-engineered features are often not accurate or comprehensive enough; (2) The linear function cannot fit the complex reward function; (3) The generalization of the reward function across different scenarios is poor.

To tackle the challenges mentioned above, deep inverse reinforcement learning (DIRL) leverages deep neural networks with enhanced fitting capabilities to represent the expert's reward function, termed the reward network [29], [30]. DIRL not only spares the trouble of Hand-engineered features but also bolsters the model's generalization by incorporating scenario information [31], [32], [33], [34]. Thus, DIRL can address the challenges posed by feature-based IRL. However, in continuous and vast solution spaces, DIRL encounters challenges in optimizing the policy corresponding to the reward network, thereby impeding convergence during training. Given this, we introduced a sampling-based DIRL approach in previous research. This method facilitates

convergence during training by discretizing the solution space and reconstructing the partition function [35], [36], [37].

B. Scenario Feature Extraction and Interpretability

Extracting scenario features from complex scenarios has long been a significant challenge, whether for human-like or conventional decision-making and planning methods. Scenario feature extraction involves understanding the scenario and identifying key information that adequately represents its characteristics. Two key factors that complicate feature extraction in multi-vehicle interaction scenarios are the high dimensionality of the scenario and the complexity of the interactions. Given the large number of vehicles in such scenarios, considering all of them can lead to information overload. To address this, most current research adopts a scenario feature extraction method similar to how human drivers perceive multi-vehicle interactions, specifically by selecting key target vehicles. This approach first ranks the vehicles by importance and then selects the key targets based on this ranking, effectively reducing the dimensionality of the scenario. After dimensionality reduction, the ego vehicle can focus its attention on the key targets, treating their information as the relevant scenario features, and use it to plan safe and feasible trajectories.

Currently, algorithms for ranking traffic vehicles based on their importance can be broadly classified into two categories. The first category is heuristic ranking algorithms, which rely on experience or rules to determine the ranking. For example, some heuristic methods rank vehicles based on the time required for them to reach the ego vehicle's expected path. The second category consists of supervised learning-based ranking algorithms, which require data labeled with importance rankings to train the ranking model. Compared to heuristic algorithms, supervised learning-based ranking algorithms generally produce more accurate ranking results [38]. However, obtaining training data with importance ranking labels is a significant challenge, which limits the application of supervised learning-based ranking algorithms.

Even when traffic vehicles can be accurately ranked by importance, relying solely on the ranking results makes it difficult to establish an interaction model among vehicles and plan reasonable trajectories. Therefore, some studies have proposed end-to-end scenario feature extraction methods, which directly extract scenario features from high-dimensional scenario information. Currently, most end-to-end scenario feature extraction methods utilize convolutional neural networks (CNN) to extract scenario features from high-dimensional data [39]. However, due to the black-box nature of CNN, these methods lack the necessary interpretability.

Interpretability has become a focal point in autonomous driving research. Currently, the interpretability of decision-making and planning results in autonomous driving can be categorized into five main forms: Attention Visualization [40], [41], Interpretable Tasks [42], [43], [44], Motion Fields [45], Linguistic Interpretability [46], [47], and Uncertainty Modeling [48], [49]. Among these methods, Linguistic Interpretability and Uncertainty Modeling require additional supervised learning models to generate interpretability outputs,

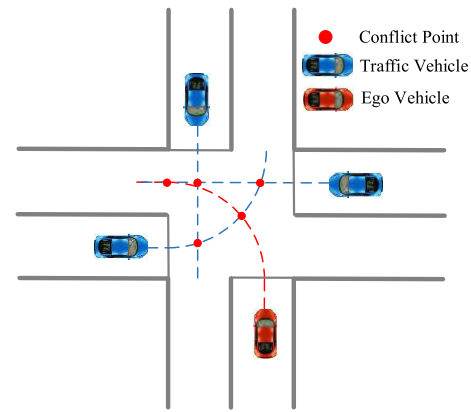


Fig. 2. The multi-vehicle interaction scenario.

which increases their technical complexity. On the other hand, Interpretable Tasks and Motion Fields are typically applied to end-to-end autonomous driving models that directly process visual information. This study focuses on human-like decision-making and planning algorithms based on structured scenario information, which includes details about traffic participants and road attributes. Given this context, Attention Visualization is the most suitable choice for enhancing the interpretability of decision-making results in the proposed human-like decision-making and planning algorithm.

Recent studies have shown that Transformer attention blocks can be used to visualize the attention weights assigned by the ego vehicle to different traffic participants [50], [51]. Furthermore, decision-making and planning networks based on Transformers can effectively generate attention weight distributions through self-supervised learning. These findings demonstrate that using Transformer attention blocks for attention visualization is a viable approach. However, most existing research has focused on behavior cloning methods, and no literature has yet verified whether Transformer-based deep inverse reinforcement learning methods can leverage their attention mechanisms for visualization. To address this research gap, we propose a Transformer-based deep inverse reinforcement learning approach, which visualizes the attention weights assigned by the ego vehicle to various traffic participants using the Transformer attention mechanism.

III. METHODOLOGY

A. Problem Formulation

Extracting scenario features serves to reduce scenario dimensionality, thereby averting input overload in the decision-making and planning model. In multi-vehicle interaction scenarios, potential conflict points may arise between the trajectory of the ego vehicle and those of traffic vehicles, as depicted in Fig. 2.

Consequently, comprehensive consideration of interactions with traffic vehicles is imperative for the safe navigation of the ego vehicle. Moreover, besides interacting with the ego vehicle, these other vehicles also engage in interactions with each other. As a result, interactions in multi-vehicle scenarios are intricately complex and challenging to model. Game theory is a mechanism-based method for modeling interactions.

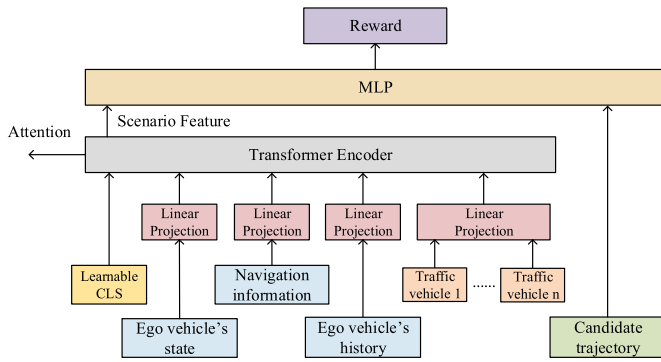


Fig. 3. The reward network of transformer-based deep inverse reinforcement learning.

However, due to the complex direct or indirect interactions among vehicles in multi-vehicle scenarios, determining the participants in game theory-based modeling is arduous and faces the challenge of the curse of dimensionality [52]. Consequently, mechanism-based methods like game theory struggle to effectively model interactions in multi-vehicle scenarios, leading to difficulties in extracting scenario features.

To this end, we propose a human-like decision-making and planning method with Transformer-based deep inverse reinforcement learning. In this method, high-dimensional scenario information is directly fed into the Transformer encoder, which automatically models the interaction among vehicles based on scenario information and transforms these interactions into scenario features from the perspective of the ego vehicle. Additionally, our proposed method provides interpretability to decision-making and planning results by computing the attention allocated by the ego vehicle to each traffic vehicle in the scenario. This attention quantifies the importance of each traffic vehicle to the decision of the ego vehicle. Subsequent sections will delve into the implementation details of this human-like decision-making and planning method.

B. Transformer-Based Deep Inverse Reinforcement Learning

In response to the challenge of deep inverse reinforcement learning struggling to converge due to the difficulty in finding the optimal policy corresponding to the reward network in continuous and extensive solution spaces, we previously proposed a sampling-based maximum entropy deep inverse reinforcement learning method [35]. The inference process of this method can be divided into three stages: sampling, evaluation, and selection. Specifically, (1) a trajectory sampler generates a candidate trajectory set based on driving scenarios; (2) the reward network evaluates the reward value of each candidate trajectory; (3) the trajectory with the highest reward is selected as the planning result. Similarly, our proposed interpretable human-like decision-making and planning method with Transformer-based deep inverse reinforcement learning also employs sampling-based deep inverse reinforcement learning. In sampling-based DIRM, the reward network and the method of sampling candidate trajectories are crucial.

The framework of the reward network for the proposed method is illustrated in Fig. 3. Its inputs consist of two components: scenario information and the candidate

trajectory. In the proposed method, the state of the ego vehicle or the traffic vehicle is represented as object-level representations, which can be described as a vector $\mathbf{o}_{i,t} = [x_{i,t}, y_{i,t}, \theta_{i,t}, vx_{i,t}, vy_{i,t}]^T$. The attributes of $\mathbf{o}_{i,t}$ include the position of the vehicle $(x_{i,t}, y_{i,t})$, heading $\theta_{i,t}$, and the velocity of the vehicle $(vx_{i,t}, vy_{i,t})$. Here, i denotes the vehicle ID, and t indicates the timestamp of the state. The navigation information of the ego vehicle in Figure 3 covers the trajectory points within the future 50 meters, with each point comprising the position and heading of the ego vehicle, spaced 0.5 meters apart. The ego vehicle's history comprises trajectory points from the previous 1 second, with each point containing the ego vehicle's position and heading, spaced 0.1 seconds apart. Before being fed into the Transformer encoder, the states of both the ego vehicle and the traffic vehicles, along with the ego vehicle's navigation information and its history, are individually embedded as vectors of equal length via their respective linear projections.

C. Interpretability: Attention Allocated to Each Traffic Vehicle

In the proposed reward network, the Transformer encoder takes as input the state information of each surrounding vehicle along with that of the ego vehicle. After processing, the encoder produces an output for each input. Directly using the output corresponding to a specific input as the scenario feature would limit the extracted feature to the perspective of that particular input, failing to encompass the global context. For example, using the output associated with a single surrounding vehicle would only capture the scenario from that vehicle's viewpoint, lacking an overall understanding. To overcome this limitation, we introduce a [CLS] token as an additional input to the Transformer encoder. After processing, the output associated with the [CLS] token is used as the scenario feature. This design ensures that the extracted features capture a comprehensive and global representation of the scenario. Hence, the output linked to the [CLS] token is employed as the scenario feature. Moreover, leveraging the attention scores of the [CLS] token on each traffic vehicle within the Transformer encoder, we compute the ego vehicle's attention allocation to every traffic vehicle during decision-making and planning. This attention quantifies each traffic vehicle's importance to the ego vehicle's decisions, thereby amplifying the interpretability of both the scenario feature and the planning outcomes. The computation method for the attention allocated to a traffic vehicle is as follows:

$$A_i = \frac{1}{M} \sum_{l=1}^M \frac{\alpha_{CLS,i}}{\sum_{j=1}^n \alpha_{CLS,j}}, \quad (1)$$

where M denotes the number of heads in the multi-head self-attention model; $\alpha_{CLS,i}$ represents the attention score of the [CLS] token on the i -th traffic vehicle; and n denotes the number of traffic vehicles in the scenario.

D. Selecting Trajectories as Vehicle Actions

During the interaction, the ego vehicle maintains its navigation information unchanged and solely adjusts its longitudinal

trajectory. In such cases, the SL coordinate system of the ego vehicle can be constructed accordingly. In this coordinate system, the ego vehicle's longitudinal trajectory comprises longitudinal trajectory points, including longitudinal position S , velocity v , and acceleration a . The longitudinal vehicle dynamics model can be represented as

$$\begin{cases} \dot{S} = v \\ \dot{v} = a. \end{cases} \quad (2)$$

According to the longitudinal vehicle dynamic model given by Formula (2), the longitudinal trajectory of the vehicle can be determined as long as the acceleration sequence within the planning horizon is provided. To reduce the number of candidate trajectories for the ego vehicle, the acceleration of the ego vehicle is set to a constant value within the planning horizon. The range of acceleration sampling is $[-5, 3]$ m/s². The sampling interval for acceleration is an adjustable hyperparameter. The selection of the sampling interval needs to balance accuracy and computational cost. A smaller sampling interval leads to higher coverage of candidate trajectories but also results in higher computational cost. To ensure the adaptability of the algorithm to different scenarios, the reward network evaluates the reward values of the candidate trajectories for the ego vehicle based on the scenario features, as illustrated in Fig. 3.

IV. TRAINING THE TRANSFORMER-BASED DEEP INVERSE REINFORCEMENT LEARNING MODEL

A. Training Algorithm

In sample-based maximum entropy IRL, the probability of the trajectory τ being selected is given by

$$P(\tau|\theta) \approx \frac{e^{R(\tau|\theta)}}{\sum_{\tau_i \in \phi} e^{R(\tau_i|\theta)}}, \quad (3)$$

where ϕ denotes the set of sampled candidate trajectories. All trajectories in ϕ have the same initial state as the trajectory τ .

In order to enhance safety while ensuring imitation, the objective function of the proposed method incorporates both imitation and safety terms, given by

$$J(\theta) = J_{\text{imitation}}(\theta) + \omega \cdot J_{\text{safety}}(\theta), \quad (4)$$

where $J_{\text{imitation}}$ represents the imitation term, unique to the objective function of conventional DIRL; J_{safety} represents the safety term, with ω indicating its weighting coefficient.

The imitation term $J_{\text{imitation}}$ can be described by

$$J_{\text{imitation}}(\theta) = \sum_{\tau_+^e \in E} \log P(\tau_+^e|\theta), \quad (5)$$

where E signifies the expert positive demonstrations; τ_+^e denotes the expert positive trajectories; $P(\tau_+^e|\theta)$ stands for the probability of selecting the expert positive trajectories τ_+^e . Positive demonstrations refer to correct behaviors made by experts that should be imitated or learned. The imitation term J_{safety} can be written as

$$J_{\text{safety}}(\theta) = \sum_{k=1}^K \log \left(1 - \sum_{\tau_-^e \in E_-^k} P(\tau_-^e|\theta) \right), \quad (6)$$

Algorithm 1 Training PN-DIRL

Input: Positive and negative demonstration dataset

Output: Transformer-based reward network

```

1: Initialize parameters of transformer-based reward network  $\theta$ 
2: for  $Epoch \leftarrow 1$  to  $Max\_Epoch$  do
3: Randomly partition the positive demonstrations into  $N$ 
   sets of equal size  $K$ , serving as mini-batches  $\kappa_1, \kappa_2, \dots, \kappa_n$ 
4: for  $n \leftarrow 1$  to  $N$  do
5:   for  $k \leftarrow 1$  to  $K$  do
6:     Compute the reward  $R(\tau_{k+}^e|\theta)$  for the positive
       demonstration  $\tau_{k+}^e$  in the mini-batch  $\kappa_n$ 
7:     Generate the candidate joint trajectory set  $\phi^k$ 
8:     for  $\tau_i^e$  in  $\phi^k$  do
9:       Compute the reward of joint trajectories  $\tau_i^e$ 
10:    end for
11:    Compute the partition function
12:    Sample negative demonstration set  $E_-^k$  with the
       same scenario state as the positive demonstration  $\tau_{k+}^e$ 
13:    for  $\tau_-^e$  in  $E_-^k$  do
14:      Compute the reward and probability of  $\tau_-^e$ 
15:    end for
16:    Compute the probability sum of all negative
       demonstrations
17:  end for
18:  Compute the objective function  $J(\theta)$ 
19:  Compute the gradient  $\nabla_{\theta} J(\theta)$ 
20:  Update the parameters  $\theta$  by the gradient rise algorithm
21: end for
22: end for
23: Output Transformer-based reward network

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where K denotes the quantity of scenarios within the training dataset; E_-^k represents the dataset encompassing all negative demonstrations in the k -th scenario; τ_-^e signifies trajectories in the negative demonstrations. Negative demonstrations refer to incorrect driving behaviors that could lead to safety incidents and should be avoided.

The training algorithm of the proposed method is outlined in Algorithm 1. Initially, the parameters of the reward network of Transformer-based DIRL are initialized. Within each episode, candidate trajectories for the ego vehicle are sampled. Subsequently, the reward values for all candidate trajectories, positive demonstrations, and negative demonstrations are computed based on the current reward network. Following this, the imitation and safety terms of the objective function are computed using these reward values. Then, the objective function for training is computed based on the imitation and safety terms, and its gradient is determined. Finally, the parameters of the reward network are updated using the gradient ascent algorithm.

In the reward network, the feature dimension of the Transformer encoder is set to 16, with 4 attention heads. The encoder for the ego vehicle and surrounding vehicles' information consists of a two-layer fully connected network. The input features are first mapped to 8 dimensions through a linear transformation, followed by processing with the ReLU

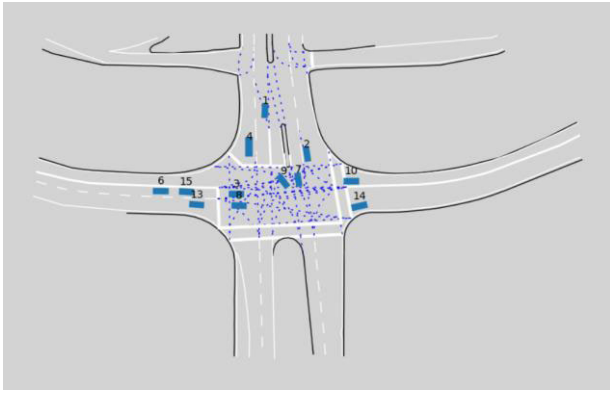


Fig. 4. DR_USA_Intersection_MA map of the INTERACTION dataset.

activation function, and then further mapped to a 16-dimensional output. The encoder for the navigation information also uses a two-layer fully connected network. It maps the 300-dimensional input features to 16 dimensions through a linear transformation, followed by ReLU activation, and finally outputs a 16-dimensional vector. The encoder for the ego vehicle's history is similarly structured with two fully connected layers. The 30-dimensional input features are mapped to 8 dimensions through a linear transformation, processed with the ReLU activation function, and then mapped to a 16-dimensional output. The multi-layer perceptron in the reward network consists of two fully connected layers. It first maps the 16-dimensional input features to 8 dimensions through a linear transformation, processes them with the Sigmoid activation function, and then maps the result to a 1-dimensional output.

Throughout the training process, a learning rate of 0.001 is designated; the mini-batch size is configured to 512; the training iterations amount to 100,000; and the weighting coefficient ω for the safety term is defined as 500.

B. Human Driving Data

To train and validate the proposed method, multi-vehicle interaction data from human drivers is extracted from the DR_USA_Intersection_MA map of the INTERACTION dataset. INTERACTION is a dataset specifically designed to record trajectory data of pedestrians and vehicles in scenarios involving intense interaction [53]. This dataset includes records of the positions and velocities of pedestrians and vehicles, as well as the heading, length, and width of vehicles. The information is recorded at a frequency of 10 Hz. The DR_USA_Intersection_MA map is depicted in Fig. 4.

Each training sample consists of three components: scenario information, a positive demonstration, and multiple negative demonstrations. Following the algorithmic framework outlined in Algorithm 1, we extracted the ego vehicle's state, its history, navigation information, and the states of all traffic vehicles from each frame of the dataset as scenario information. The actions taken by human drivers in this context are considered as the positive demonstration, while actions leading to collisions are regarded as negative demonstrations. Consequently, each scenario contains only one positive demonstration, but multiple negative demonstrations may exist. Across the 22 data

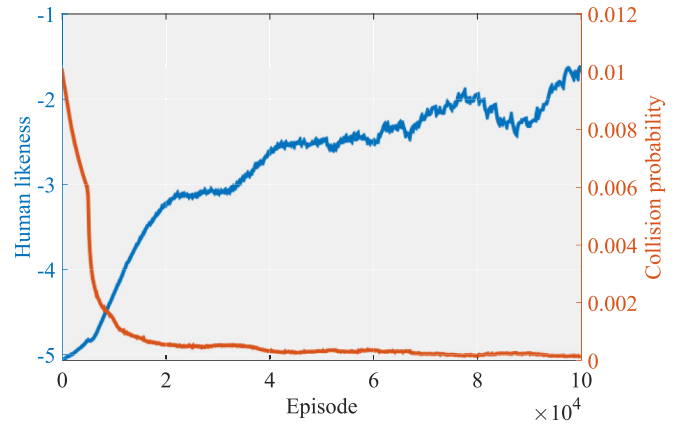


Fig. 5. Evolution process of the model.

files of the DR_USA_Intersection_MA map, over 500,000 training samples are extracted, with 70% utilized as the training dataset and the remaining 30% as the test dataset.

V. EXPERIMENTS AND DISCUSSIONS

A. Evaluation of Imitation and Safety

Imitation (human-likeness) refers to the extent to which the driving behavior of an autonomous vehicle resembles that of a human driver. This performance is generally quantified by comparing the similarity between the trajectories of both. Common metrics for this include Average Displacement Error (ADE) and Final Displacement Error (FDE). For algorithms based on probabilistic models, the likelihood of human driver behavior is often used as a metric for quantification. Safety refers to the autonomous vehicle's ability to avoid collisions in various scenarios, and is typically quantified by the collision rate.

We train the proposed model using the data collected in Section IV, and the evolution of the model's imitation and safety is illustrated in Fig. 5. The blue curve represents the evolution of the model's imitation, while the red curve represents the evolution of the model's safety. From the trend of the model's imitation, it can be observed that as the model evolves, the log-likelihood of expert positive demonstrations rises from an initial -5.07 to -1.64 , indicating a gradual improvement in the model's imitation during training, eventually converging. Regarding the trend of the model's safety, it can be seen that as the model evolves, the collision probability decreases from an initial 1.00% to 0.01%. This indicates a significant enhancement in the safety of the model after training. The training results demonstrate that the proposed method can simultaneously improve the imitation and safety of the model.

The imitation-related and safety-related metrics are used to evaluate the model's performance in terms of imitation and safety. For safety metrics, this study employs the log-sim testing method, a widely recognized approach in both academic and industrial domains for assessing imitation performance [54], [55], [56]. In log-sim testing, the ego vehicle is controlled by the model instead of following its original trajectory from the dataset, while the other traffic vehicles adhere to their respective original trajectories recorded in

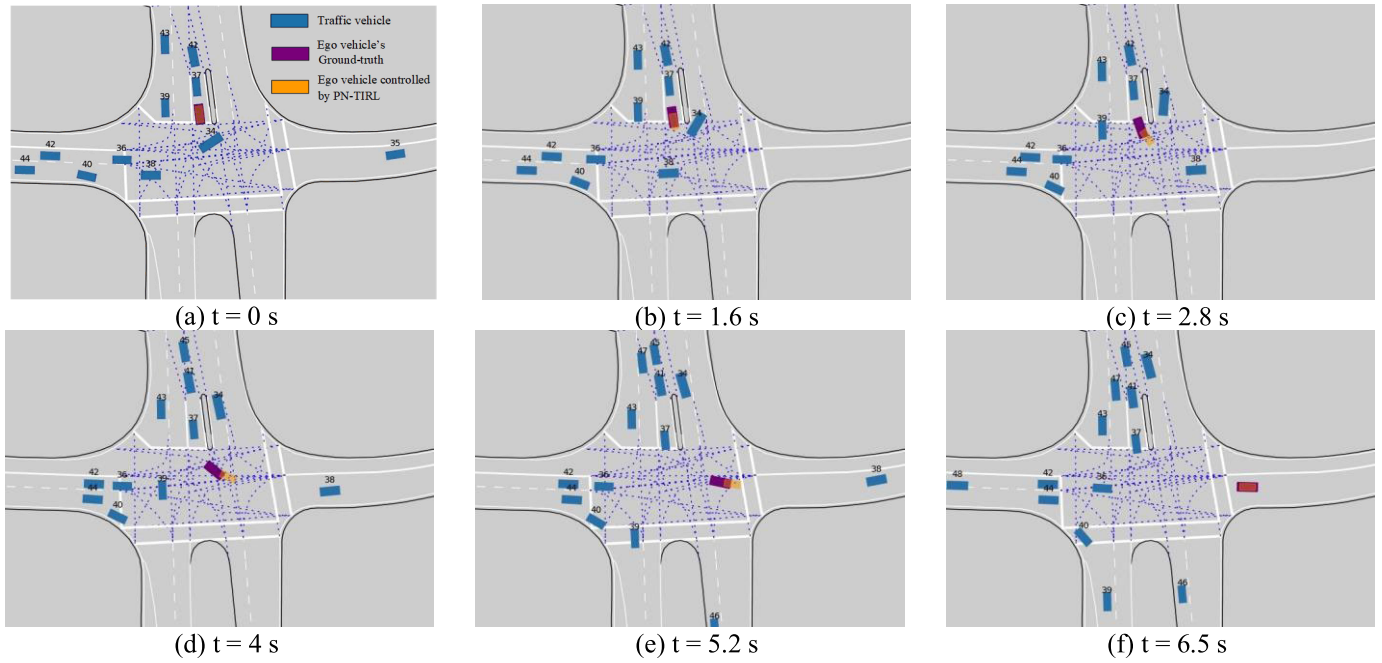


Fig. 6. PN-TIRL's performance in the unsignalized intersection.

the dataset. For imitation metrics, an interactive simulation environment is used. In this environment, the initial state is set according to the scenarios recorded in the dataset, and the subsequent motion of traffic vehicles is driven by the traffic model in the simulation software PanoSim®. This enables interaction capabilities between the ego vehicle and other traffic vehicles.

The imitation-related metrics are as follows:

(1) Root Mean Square Error (RMSE) of acceleration: The RMSE between the acceleration computed by the model and that of the expert positive demonstrations.

(2) Average Displacement Error (ADE): The average displacement error between the planned trajectory and the trajectory of the expert positive demonstrations.

(3) Final Displacement Error (FDE): The displacement error at the endpoint between the planned trajectory and the trajectory of the expert positive demonstrations.

The safety-related metrics are as follows:

(4) Collision Rate: The rate of scenarios resulting in collisions during testing among all testing scenarios.

We conducted a quantitative evaluation of three models in a closed-loop simulation: behavior cloning (BC), the Transformer-based deep inverse reinforcement learning model trained only on expert positive demonstrations (TIRL), and the Transformer-based deep inverse reinforcement learning model trained on both positive and negative demonstrations (PN-TIRL). As shown in TABLE I, the proposed PN-TIRL outperforms both BC and TIRL in terms of imitation and safety. While the training objectives of BC and TIRL focus solely on improving the model's imitation, PN-TIRL is designed to enhance both imitation and safety. The experimental results demonstrate that PN-TIRL achieves significant safety improvements without compromising imitation performance, even with its dual training objectives.

TABLE I
IMITATION AND SAFETY COMPARISON BETWEEN
BC, TIRL AND PN-TIRL

Model	Imitation			Safety
	RMSE	FDE	ADE	Collision Rate
BC	0.35	3.53	1.75	6.78%
TIRL	0.14	3.16	1.28	4.93%
PN-TIRL (Ours)	0.09	2.32	0.94	0.07%

For a more vivid demonstration of PN-TIRL's effectiveness, we visualize its planning results at an unsignalized intersection from the INTERACTION dataset, as shown in Fig. 6. The experimental approach is based on the log-sim method. In Fig. 6, the purple vehicles represent the ground-truth positions of the ego vehicle in the dataset, while the yellow vehicles are controlled by the model. The blue vehicles represent traffic vehicles, following their respective original trajectories from the dataset. In this scenario, the ego vehicle is preparing to turn left from the upper intersection to the right intersection. Until $t = 1.6$ s, the ego vehicle remains stationary within the stop line to yield to traffic vehicle 34. After traffic vehicle 34 passes, the ego vehicle starts moving. From Fig. 6 (b), it can be observed that at $t = 1.6$ s, there is a potential conflict between the ego vehicle and traffic vehicle 38. Eventually, the ego vehicle adopts a yielding strategy for traffic vehicle 38, as shown in Fig. 6 (c). The experimental results demonstrate that the planned trajectory by the model closely aligns with the ego vehicle's ground-truth trajectory. This indicates that the proposed human-like decision-making and planning method can learn driving behaviors from human

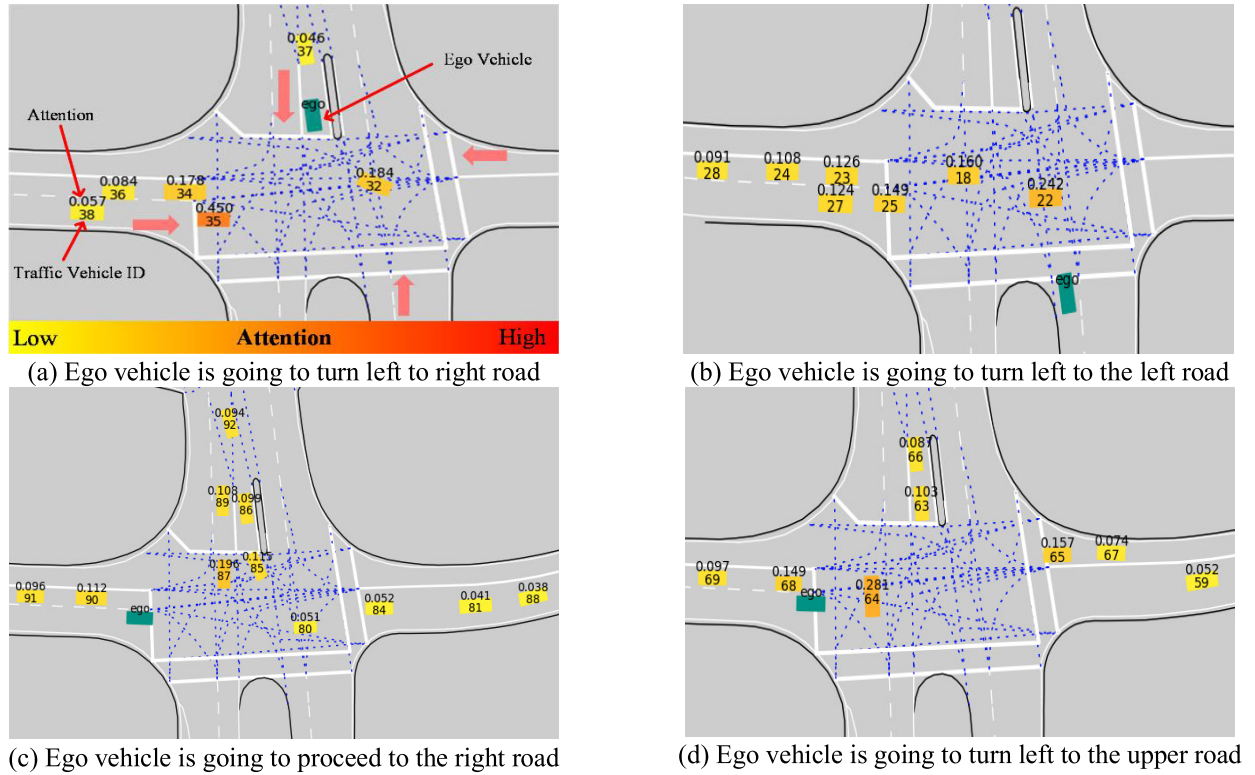


Fig. 7. Attention allocated by the ego vehicle to each traffic vehicle. The blue vehicle represents the ego vehicle, while vehicles of other colors depict traffic vehicles. The darkness of the color of traffic vehicles indicates the level of attention they receive, with darker colors suggesting higher attention allocated by the ego vehicle. The numbers inside the traffic vehicles represent their IDs, while the numbers above them indicate their attention. The arrows in Fig. 7(a) indicate the directions of the lanes.

driving data, exhibiting a driving style and proficiency similar to that of human drivers in multi-vehicle interaction scenarios.

B. Evaluation of Interpretability

Interpretability refers to the degree to which human users can understand the decision-making results of the model. Since understanding is a subjective process, it is typically assessed through the subjective feedback of the driver. Our proposed method enhances the interpretability of decision-making and planning results by computing the attention allocated by the ego vehicle to each traffic vehicle in the scenario. This attention quantifies the importance of each traffic vehicle to the decision of the ego vehicle. We determine this attention by utilizing the attention scores of the [CLS] token on each traffic vehicle within the Transformer encoder, as shown in Formula (1). To validate the interpretability of the proposed method, we present the attention allocation results of the model in various scenarios, as depicted in Fig. 7.

In the scenario shown in Fig. 7(a), the ego vehicle is going to turn left onto the right road. Three traffic vehicles receiving the highest attention are traffic vehicles 35, 32, and 34, respectively. Traffic vehicles 35 and 34 are going to proceed onto the right road, potentially posing a collision risk to the ego vehicle. Traffic vehicle 32, similar to the ego vehicle, is also going to turn left onto the right road but is positioned ahead of the ego vehicle. To prevent a collision, the ego vehicle is focusing its attention on these three traffic vehicles.

In the scenario depicted in Fig. 7(b), the ego vehicle is going to turning left, while all traffic vehicles are going to proceed from the left road to the right road. Traffic vehicles 22 and 18 are the first ones to interact with the ego vehicle, receiving the highest attention. Additionally, the farther the traffic vehicle is from the ego vehicle, the lower its attention. In the scenario illustrated in Fig. 7(c), the ego vehicle is going to proceed to the right road. There is a potential collision risk between the ego vehicle and traffic vehicles 87 and 85. Since the ego vehicle needs to interact with traffic vehicle 87 first, it allocates the highest attention to traffic vehicle 87. Additionally, considerable attention is allocated to traffic vehicle 90. Despite being behind the ego vehicle, the potential risk of collision if it turns left warrants attention. This aligns with the driving habits of human drivers.

In Fig. 7 (d), the ego vehicle is turning left onto the upper lane. Traffic vehicles 64, 65, 68, and 63 pose collision risks, receiving the highest attention from the ego vehicle. Despite being farther from the ego vehicle, traffic vehicle 65 still receives significant attention, indicating that the ego vehicle considers collision risks rather than just distance, aligning with human drivers' attention allocation habits.

The attention allocation results across the aforementioned scenarios indicate a remarkable similarity between the model's attention allocation and that of human drivers. The experimental findings suggest that the model's attention allocation is reasonable and provides clear interpretability for the model's decision-making and planning results.

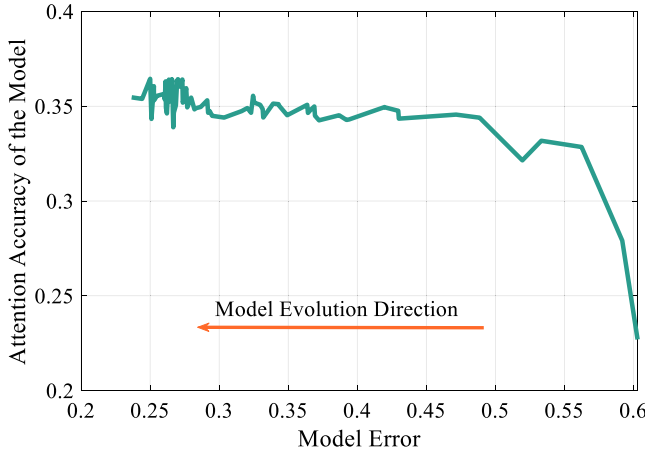


Fig. 8. Correlation between model error and attention.

To quantify the accuracy of the attention output from the model, we introduced the metric “Attention Accuracy”, which is defined as the ratio of attention allocated to the three most crucial traffic vehicles labeled by human drivers to the total attention across all traffic vehicles. The higher this ratio is, the more precise the model’s attention allocation is. Attention accuracy is given by

$$P_{att} = \frac{A_1 + A_2 + A_3}{\sum_{i=1}^n A_i}, \quad (7)$$

where P_{att} is attention accuracy of the model; A_1 , A_2 , and A_3 represent the attention allocated to the three most crucial traffic vehicles labeled by human drivers; $\sum_{i=1}^n A_i$ represents the sum of the attention for all traffic vehicles.

Fig. 8 illustrates the correlation between model error and attention accuracy, revealing the critical role of attention mechanisms in improving model performance. As shown in Fig. 7, the model’s attention mechanism accurately identifies key traffic participants with potential conflict risks in complex scenarios, mirroring human-like attention allocation patterns. This capability supports collision avoidance during interactions and validates the interpretability of the model’s attention weight distribution. From the experimental results in Fig. 8, it is evident that as the model error decreases, attention accuracy improves significantly. This suggests that a more precise attention mechanism effectively enhances model performance. The diverse scenarios in Fig. 7 further reveal how the model allocates attention in complex traffic interactions. For instance, in the four scenarios depicted, the model assigns higher attention weights to traffic vehicles with potential conflict risks, thereby increasing both human-likeness and safety. The connection between attention accuracy and error reduction, as shown in Fig. 8, can be interpreted as the model progressively learning human-like attention allocation patterns by minimizing errors. A more accurate attention mechanism, in turn, supports further error reduction.

Additionally, it is worth noting that the attention accuracy is not directly optimized in the training objective but emerges as an intermediate output of the reward network. This result,

aligned with the attention allocation patterns observed in Fig. 7, demonstrates that the model can autonomously extract meaningful attention strategies from driving data without labeled supervision. This not only validates the learning capability of the attention mechanism but also confirms that the model can indirectly acquire accurate attention through error optimization, driving continuous performance improvement.

C. Hardware-in-the-Loop Experiment

To verify the effectiveness of the proposed method in integrating control modules and operating within a hardware-in-the-loop (HIL) environment, we develop longitudinal and lateral control algorithms for the autonomous vehicle and establish a hardware-in-the-loop experimental platform. The HIL experimental platform utilized PanoSim®, an autonomous driving simulation software, as its simulation environment.

Firstly, to track the lateral and longitudinal trajectories planned by the proposed algorithm, we designed lateral trajectory tracking and longitudinal trajectory tracking algorithms based on the model predictive control (MPC). The state-space representation of the lateral trajectory tracking algorithm is designed as

$$\begin{bmatrix} \dot{y} \\ \dot{v} \\ \dot{\theta} \\ \dot{\omega}_r \end{bmatrix} = \begin{bmatrix} 0 & 1 & u & 0 \\ 0 & -\frac{2(k_1+k_2)}{mu} & 0 & \frac{2(k_2b-k_1a)}{mu} - u \\ 0 & 0 & 0 & 1 \\ 0 & \frac{2(k_2b-k_1a)}{I_z u} & 0 & -\frac{2(k_1a^2+k_2b^2)}{I_z u} \end{bmatrix} \begin{bmatrix} y \\ v \\ \theta \\ \omega_r \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{2k_1}{m} \\ 0 \\ \frac{2k_1a}{I_z} \end{bmatrix} \cdot \delta, \quad (8)$$

where a represents the distance from the front wheel to the center of mass; b denotes the distance from the rear wheel to the center of mass; δ represents the front wheel steering angle; k_1 and k_2 indicate the lateral stiffness of the front and rear wheels, respectively; u and v indicate the longitudinal and lateral velocities of the vehicle, respectively; ω_r denotes the yaw angular velocity of the vehicle; m represents the total mass of the vehicle; I_z indicates the rotational inertia of the vehicle around the Z-axis; y denotes the lateral position; θ indicates the heading angle.

The state-space representation of the longitudinal trajectory tracking algorithm is designed as

$$\begin{bmatrix} \dot{S} \\ \dot{v} \\ \dot{a} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1/T_x \end{bmatrix} \begin{bmatrix} S \\ v \\ a \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1/T_x \end{bmatrix} a_{req}, \quad (9)$$

where S , v , and a represent the longitudinal station in the S-L coordinate system, velocity, and acceleration, respectively; a_{req} is the requested acceleration of the controller; T_x is the time constant of the first-order inertial element. According to the identification method of the first-order inertial element, the time constant T_x is equal to the time for the output of system rises to 63.2% of the final value.

The cost function of MPC is expressed as

$$J = (\eta - \eta_r)^T Q (\eta - \eta_r) + U^T R_1 U + \Delta U^T R_2 \Delta U, \quad (10)$$

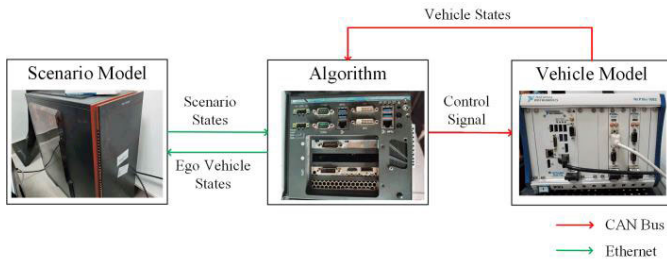


Fig. 9. The Hardware-in-the-loop system.

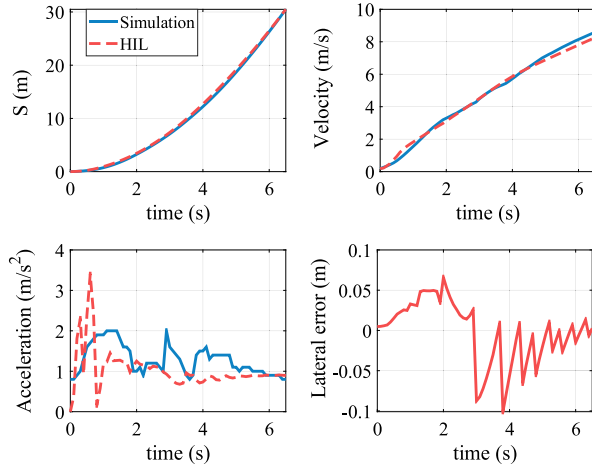


Fig. 10. The experimental results of simulation and HIL.

where η_r represents the desired trajectory; η represents the actual trajectory; U represents the control sequence; Q represents the weighting of state variables; R represents the weighting of control variables; ΔU represents the sequence of control increments. In the cost function, the first term aims to align the actual output sequence closely with the expected output sequence. The second term aims to minimize the control input, fostering control stability and reducing energy consumption in actuators. The third term focuses on minimizing control input oscillations.

The hardware-in-the-loop (HIL) experiment assesses the proposed algorithm's effectiveness and real-time performance. The entire HIL system is depicted in Fig. 9, encompassing the scenario model platform, the algorithm platform, and the vehicle model platform. Communication between the vehicle model and the algorithm is achieved via a CAN bus with a throughput of 500 Kbit/s, while communication between the scenario model and the algorithm utilizes Ethernet.

The quantitative results in HIL depicted in Fig. 10 reveal that the actual trajectory of the ego vehicle closely aligns with the planned trajectory, with lateral error consistently kept within 10 cm. This indicates that the longitudinal and lateral control algorithms effectively fulfill the demands of autonomous driving. It is evident that the proposed method remains effective in integrating control modules and operating within a HIL environment.

VI. CONCLUSION

We propose a Safe and Interpretable Human-like Decision-Making and Planning method using Transformer-based Deep

Inverse Reinforcement Learning to address the critical challenges of safety and interpretability in autonomous driving. The method enhances safety by training on both positive and negative demonstrations, enabling the model to learn desirable driving behaviors and identify actions to avoid, thereby minimizing collision risks. By integrating a Transformer Encoder into the reward network, our approach effectively extracts key scenario features and computes the attention weights the ego vehicle allocates to surrounding traffic participants, ensuring clear interpretability of the planning results.

We train and validate the proposed method using human driving data from multi-vehicle interaction scenarios at urban intersections. Experimental results demonstrate the outstanding performance of our method in terms of imitation and safety. Additionally, the experimental results regarding interpretability show a high consistency between the attention allocation results computed by our method and those by human drivers. This indicates that the model's attention output effectively explains the importance of each traffic vehicle for decision-making and planning results, thereby enhancing the model's interpretability. Safety and interpretability have long been major challenges in human-like decision-making and planning methods. The proposed method significantly improves both the safety and interpretability of these approaches, providing a solid foundation for future research and practical applications in this field, with both academic and real-world implications.

Our focus in this paper is on validating the effectiveness of the proposed method, without considering its generalization performance across multiple maps. In the future, efforts can be directed toward designing map encoding methods and incorporating map encoding as one of the inputs to the reward network to achieve generalization across various maps. This extension would contribute to the broader applicability and robustness of the proposed approach in diverse real-world scenarios.

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